

Social Sciences - Key Terms & Definitions

Social Sciences

Construct: is an abstract concept that is specifically chosen (or 'created') to explain a given phenomenon.

- Unidimensional construct = consists of one simple concept.
- Multi-dimensional construct = consists of multiple underlying concepts.

Contingency: a future event or circumstance which is possible but cannot be predicted with certainty.

Control Variable: A variable that is not pertinent to explaining a given dependent variable, but may have some impact on the dependent variable. These variables must be controlled for in a scientific study, and are therefore called control variables.

Deductive Research: The goal of the researcher is to test concepts and patterns known from theory using new empirical data (*theory-testing*).

Dependent Variable: A variable that is explained by other variables.

Descriptive Research: is directed at making careful observations and detailed documentation of a phenomenon of interest (answers to why and how type of questions). These observations must be based on the **scientific method (i.e., logical, confirmable, repeatable, and scrutinizable)**, and therefore, are more reliable than casual observations by untrained people.

Dictionary Definition: A construct is defined in terms of a synonym (e.g., attitude may be defined as a disposition, a feeling, or an affect).

- Note that dictionary definitions are circular and not very useful for elaborating the meaning and content of that construct.

Empirical Level: is concerned with testing the theoretical concepts and relationships to see how well they reflect our observations of reality, with the goal of building better theories.

Epistemology: is the branch of philosophy dedicated to the study of knowledge, focusing on its nature, origin, scope, and justification. Here, in the context of Social Sciences, it refers to our assumption about ***the best way to study the world*** (e.g., *should we use an objective or subjective approach to study social reality?*).

Exploratory Research: is often conducted in new areas of inquiry, where the goals of the research are **(1)** to scope out the magnitude or extent of a particular phenomenon; **(2)** to generate some initial ideas about the phenomenon; or **(3)** to test the feasibility of undertaking a more extensive study regarding the phenomenon. (Examines the what,

where, and when of a phenomenon).

Functionalism: If researchers view the world as consisting mostly of social order (ontology) and hence seek to study patterns of ordered events or behaviors, and believe that the best way to study such a world is using objective approach (epistemology) that is independent of the person conducting the observation or interpretation, such as by using standardized data collection tools like surveys, then they are adopting a **paradigm of functionalism**.

Hypotheses: Is the empirical formulation of propositions, stated as a relationship between variables.

- **Example:** *"An increase in students' IQ score causes an increase in their grade point average."*

Independent Variable: A variable that explains other variables.

Inductive Research: The goal of a researcher is to infer theoretical concepts and patterns from observed data (**theory-building**).

Mediating Variable: A variable that is explained by independent variables while also explaining dependent variables.

Methodological Skills: “Know-how” (acquired throughout education).

Moderating Variable: A variable that influences the relationship between independent and dependent variables.

Natural Science: The science of naturally occurring objects or phenomena, such as light, matter, earth, celestial bodies, or the human body.

Ontology: a branch of metaphysics concerned with reality, the nature, and relations of being. Here, in the context of Social Science, ontology refers to our assumptions of **how we see the world** (e.g., does the world consist mostly of social order or constant change?)

Operational Definition: defines the construct in terms of how it will be empirically measured (e.g., the operational definition of a construct, such as temperature, must specify whether we plan to measure temperature in Celsius, Fahrenheit, or Kelvin scale).

Paradigm: A paradigm is a mental model or frame (belief system).

- Paradigms are often hard to catch because they are implicit, assumed, and taken for granted.
- **Example:** Why was a certain technology successfully implemented in one organization but failed in another?
- Social scientists might look at this question from:

- a **rational lens**: e.g., looking for a rational explanation of the problem, such as inadequate technology.
 - a **social lens**: e.g., looking at social deficiencies such as inadequate user training.
 - a **political lens**: g., looking at instances of organizational politics that may subvert the technology implementation process.
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- **Note**: Different paradigms emphasize different aspects of the same reality!

Positivism: holds that science or knowledge creation should be restricted to what can be observed or measured. Furthermore, positivism relies exclusively on theories that can be tested. This leads to empiricism and blind faith in observed data and a rejection of any attempt to extend or reason beyond observable data.

Post-Positivism: argues that one can make reasonable inferences about a phenomenon by combining empirical observations with logical reasoning. Post-Positivists view science not as certain but probabilistic (based on many contingencies), and they often seek to explore these contingencies to understand the social reality better.

A Proposition: is a tentative and conjectural relationship between constructs that is stated in a declarative form.

- **Example of a proposition**: An increase in student intelligence causes an increase in their academic achievement.

Radical Humanism: If researchers wish to understand social change using the subjective perspectives of the participants involved, then they are following a radical humanism paradigm.

Scientific Knowledge: refers to a generalized body of laws and theories to explain a phenomenon or behavior of interest that is acquired using the scientific method.

Social Science: The science of people or collections of people, such as groups, firms, societies, or economies, and their individual or collective behavior.

Social Science Research: operates at two levels: a **theoretical level** and an **empirical level**.

Theoretical Level: is concerned with abstract concepts about a natural or social phenomenon and relationships between those concepts (i.e., building “theories”).

Theoretical Skills: “Know-what” (requires years of observation and reflection, and are tacit skills that cannot be ‘taught’ but rather learned through experience).

Unit of Analysis: refers to the person, collective, or object that is the target of the investigation.

Variable: is a measurable representation of an abstract construct.